output will be distorted.

2 **(a)** **(i)** $R = \frac{V}{I} = \frac{4.5}{2.5} = 1.8\ \Omega$

(ii) $R = \frac{\rho l}{A} \rightarrow l = \frac{RA}{\rho}$.

$$l = \frac{(1.8)\left(\pi\left(0.30\times 10^{-3}\right)^2\right)}{\left(1.6\times 10^{-8}\right)} = 31.8\text{ m}$$

Number of turns $= \frac{31.8}{0.088} = 360$

(iii) number of turns per unit length $= \frac{360}{0.12} = 3000\text{ m}^{-1}$

(b) (i) One tesla is the magnetic flux density when a conductor of length 1 m carrying a current of 1 A, placed perpendicular to the magnetic field, experiences a force of 1 N.

**Unit current is not acceptable where what is required is one ampere. Very few definitions made reference to a straight wire placed normal to the magnetic field.*

(ii) $B = \mu_0 nI = \left(4\pi \times 10^{-7}\right)\left(3000\right)\left(2.5\right) = 9.4 \times 10^{-3}\ \text{T}$

(c) (i) Along the axis, $v = \left(4.0 \times 10^{7}\right)\left(\cos 30^\circ\right) = 3.5 \times 10^{7}\ \text{m s}^{-1}$

(ii) Normal to the axis, $v = \left(4.0 \times 10^{7}\right)\left(\sin 30^\circ\right) = 2.0 \times 10^{7}\ \text{m s}^{-1}$

(d) Magnetic force acting on the particle $F = Bqv$. This force provides the centripetal force required to keep the particle in a circular path. Hence, $Bqv = \dfrac{mv^2}{r}$. Thus, $r = \dfrac{mv}{Bq}$.

********Explanation must be given for the equations that were used. Merely stating the equations will not gain full credit.*

(e) $r = \dfrac{mv}{Bq} = \dfrac{\left(9.11 \times 10^{-31}\right)\left(2.0 \times 10^{7}\right)}{\left(9.425 \times 10^{-3}\right)\left(1.6 \times 10^{-19}\right)} = 0.012\ \text{m}$

The diameter of the electron helical path (starting at the axis of the solenoid) is thus 0.024 m, greater than the radius of the solenoid (0.014 m). Thus, the electron will collide with the wall of the solenoid.

3 (a) $p = \dfrac{h}{\ldots} = \dfrac{\left(6.63 \times 10^{-34}\right)}{\ldots} = 2.8 \times 10^{-24}\ \text{N s}$